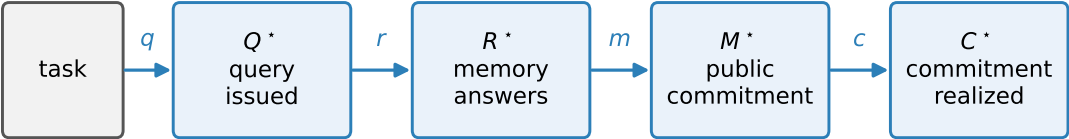
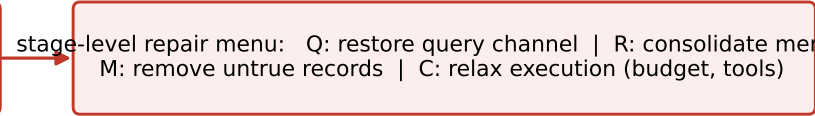
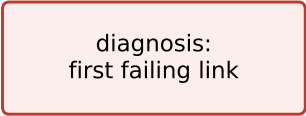


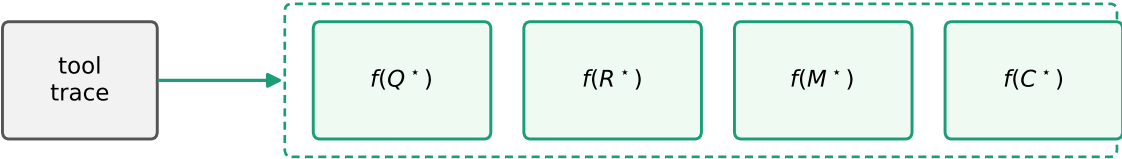
A. Nested regime (explicit query--lock chain)



nested by construction: $C^* \subseteq M^* \subseteq R^* \subseteq Q^* \Rightarrow P(C^*) = qrmc$



B. Marginal regime (no lock chain: tool traces)



independent behavioral descriptors: events not nested (M^* may occur without R^*), no product claimed